

TITLE PAGE

- **Problem Statement ID** – 4
- **Problem Statement Title** – AI Powered Behavioral Anomaly Detection for Cybersecurity
- **Theme** – Cybersecurity and Blockchain
- **PS Category** - Software
- **Student Name** - Ram Ganesh Vemula
- **Student ID** – PES1UG23CS470
- **Github Repository Link** - <https://github.com/ram77-g/Tracewell>

Tracewell (Idea Title)

Proposed Solution

- AI pipeline learns per entity behavioral baselines, not static rules
- Isolation Forest flags outliers; PyTorch GRU adds sequence-aware anomaly scoring; XGBoost classifies attack type
- Chain Linker connects alerts into full attack kill chains
- SHAP gives a plain English reason for every alert
- Dashboard with live simulation, MITRE mapping, Downloadable PDF reports

How It Addresses the Problem

- Two-tier alerting cuts noise instead of flooding analysts
- Handles cold start entities and concept drift natively
- Explains why an event was flagged, not just that it was

Innovation & Uniqueness

- Reconstructs multistep kill chains, not just single events
- Feedback loop retraines and promotes only better models
- Hybrid deep learning: GRU next-step predictor feeds a sequence anomaly score into XGBoost — full neural pattern recognition without losing SHAP explainability

TECHNICAL APPROACH

Tech Stack

MACHINE LEARNING

Python · scikit-learn (I F)
· PyTorch (GRU)
· XGBoost · SHAP

FRONTEND

React (Vite) · vis-network
(relationship graphs)

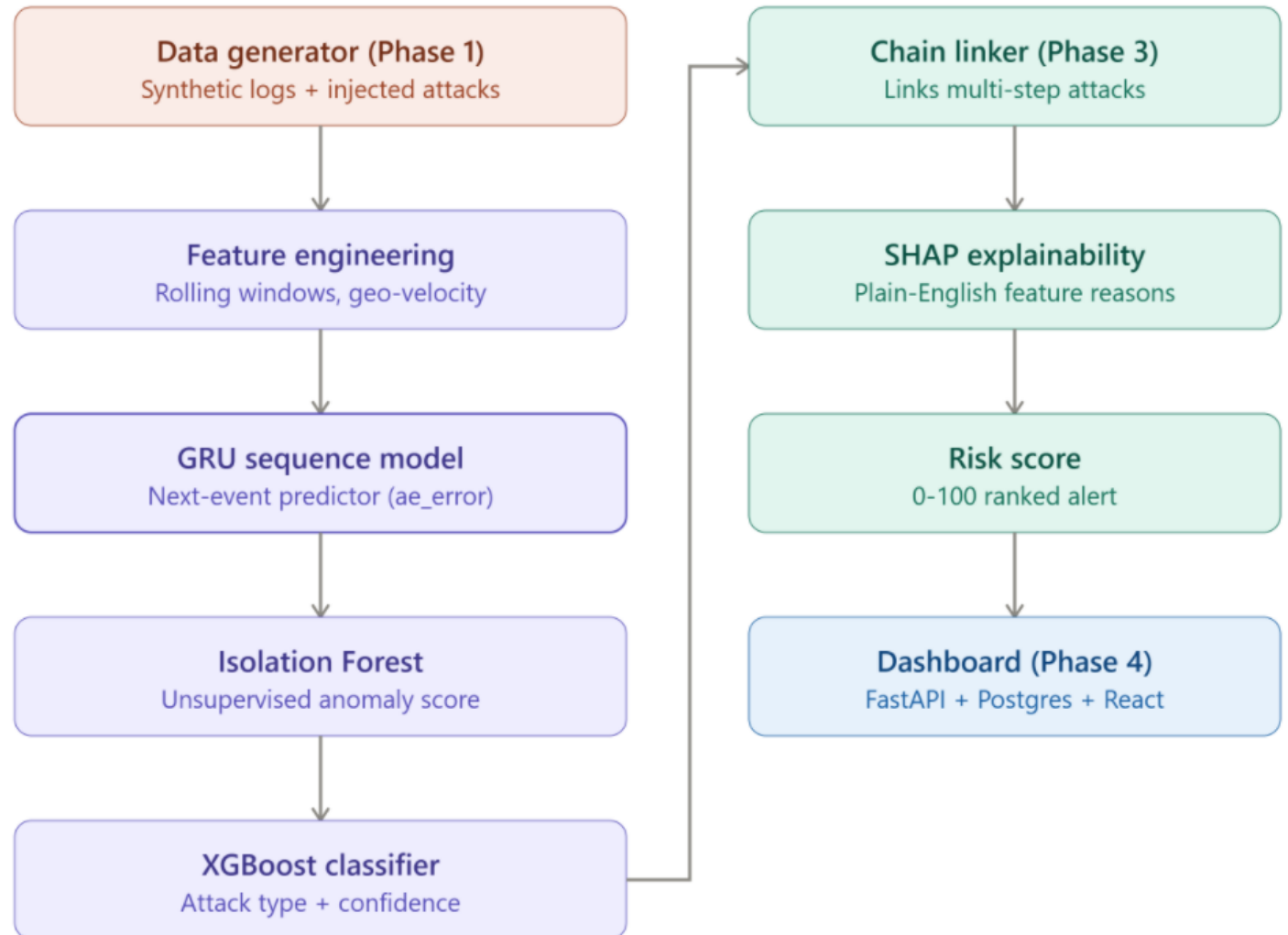
DATA GENERATION

NumPy · pandas · Faker
(synthetic access logs)

BACKEND

FastAPI · PostgreSQL ·
SQLAlchemy · WebSockets
(real-time streaming)

Methodology/Architecture



FEASIBILITY AND VIABILITY

Feasibility

- Working prototype today: live pipeline, trained models, running dashboard
- Runs on standard local hardware — no specialized infra needed
- Modular 4 phase architecture — each layer scales independently

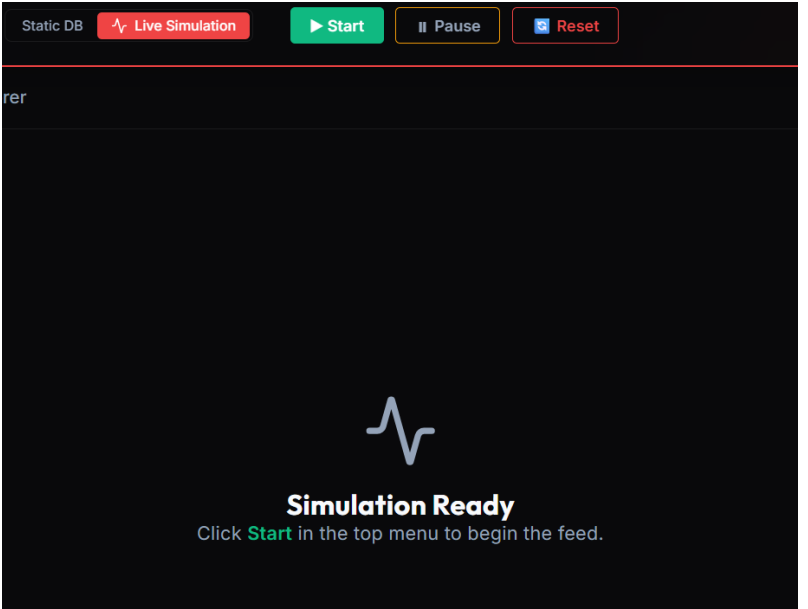
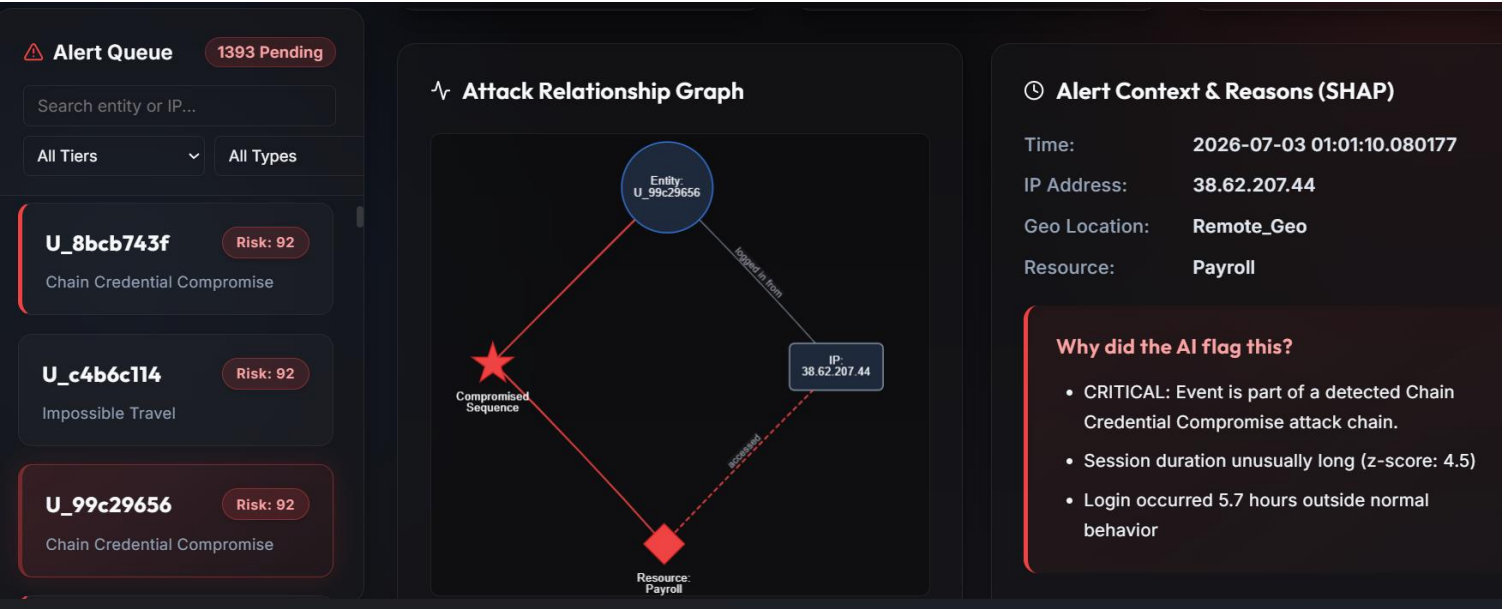
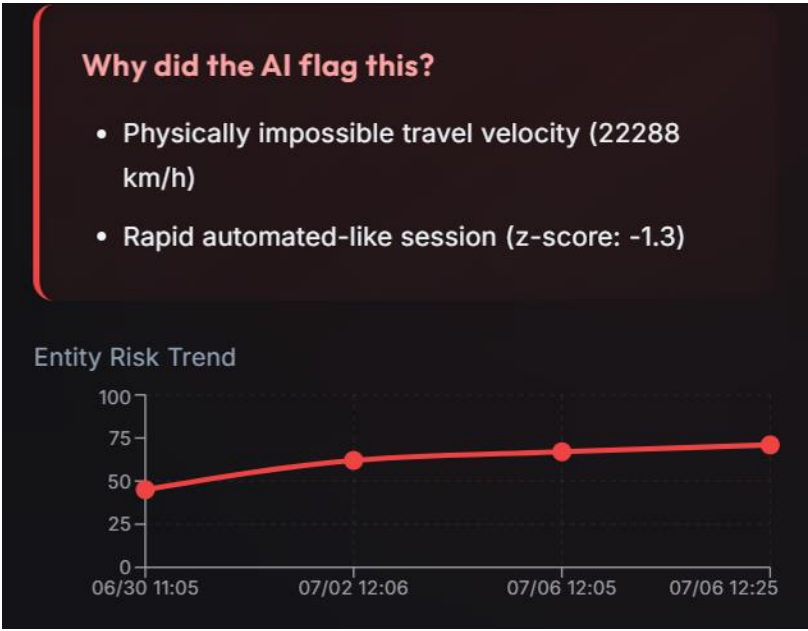
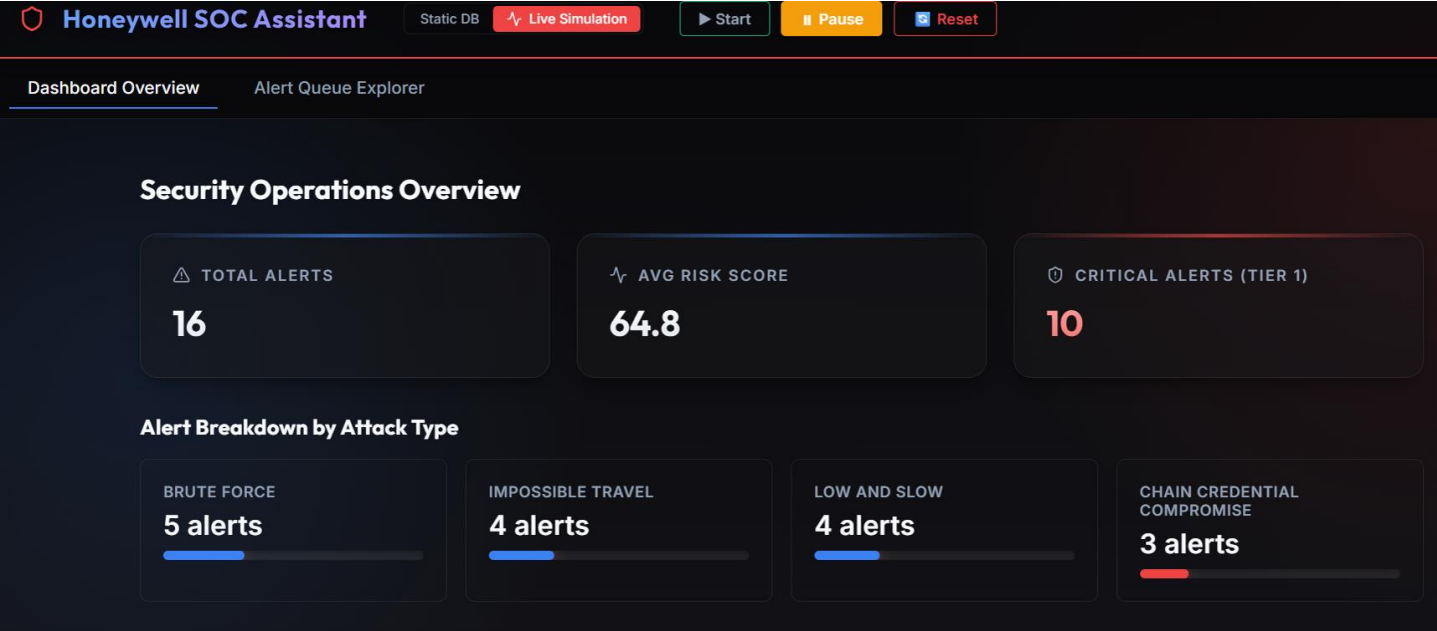
Potential Challenges & Risks

- Synthetic data can't capture every real world attack pattern
- Extreme class imbalance hurts rare attack classification
- Single node DB limits throughput at high velocity scale

Strategies to Overcome

- Feedback driven retrain pipeline improves rare class detection over time
- Two tier alerting balances precision (strict) with recall (chain-linking)
- Clear scale path: distributed streaming (Kafka) + time series DB for production

ARTIFACTS



RESEARCH AND REFERENCES

MITRE ATT&CK Framework — <https://attack.mitre.org>

Isolation Forest (Liu, Ting & Zhou, IEEE ICDM 2008) — <https://doi.org/10.1109/ICDM.2008.17>

SHAP: A Unified Approach to Interpreting Model Predictions (Lundberg & Lee, NeurIPS 2017)
— <https://arxiv.org/abs/1705.07874>

Chung, J., et al. "Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling" (2014) — <https://arxiv.org/abs/1412.3555>

UEBA Overview (CrowdStrike) — <https://www.crowdstrike.com/en-us/cybersecurity-101/identity-protection/user-and-entity-behavior-analytics-ueba>

Video: Understanding UEBA — <https://www.youtube.com/watch?v=3Z1-jCXKvO0>